

Test Plan for

SmartBreakReminder

General information

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Customer	SmartBreakReminder
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1. Introduction

1.1. General information

This document describes the methods and procedures that will be used by the QATestLab team in the functional testing process of the SmartBreakReminder web application. It is meant to be used as a manual during testing works. It describes the procedure of the testing process. The test plan is intended for project managers, product developers, and QA engineers.

The objective of the testing activities is to check functions and features of the SmartBreakReminder web application developed for modern web browsers (Chrome, Firefox, Edge, Safari) and accessible on modern Android and iOS devices via responsive design.

1.2. Purpose

This Test Plan document for the GymTracker project supports the following objectives:

- Identify existing project information and software components to be tested.
- Recommendation and description of the testing strategies to be employed.
- Identify required resources and provide a test effort estimate.
- List the test project deliverable elements.

The results of test execution will be sent to the customer as reports. All found bugs will be tracked using the Jira bug tracker.

2. Scope of project

2.1. Scope of web application

Testing of SmartBreakReminder web application is in scope. The following components and functions would be tested:

1. User Registration (via email with JWT authentication)
2. Login and password recovery
3. Workout template creation, editing and deletion
4. Exercise logging (weight, repetitions, sets, time)
5. Training session recording and management
6. Progress analytics — charts, statistics, volume calculation
7. Personal Records (PR) tracking
8. User profile: view, edit, update

2.2. Scope of mobile (responsive) testing

Testing of the responsive/mobile layout is in scope. The following will be tested:

1. User registration and login on mobile browsers
2. Navigation and UI layout on small screens (smartphones, tablets)
3. Workout logging on mobile devices
4. Charts and analytics display on mobile
5. Responsive design correctness across various screen sizes
6. Performance and load time on mobile browsers

2.3. Scope of Admin part

Testing of the Admin panel is in scope. The following components and functions would be tested:

1. Admin login and authentication
2. Exercise catalog management (CRUD: create, read, update, delete)
3. User account management (view and manage users)
4. System statistics and usage reports
5. Data integrity verification

3. Work plan

The parties agreed to follow the next work plan:

1. Test plan preparation
2. Test plan approval
3. Functional testing and bug reporting
4. Daily reports preparation
5. Final report preparation

4. Test Plan and Strategy

4.1. Functional testing

The objective of functional testing is to make sure that the whole SmartBreakReminder web application works according to the requirements, and no significant errors appear. Functional testing is the most substantial part of software testing. It involves checking different aspects of the system. A software product must pass all the planned tests. Only in this case its quality can be assured.

Test Objective: Ensure proper target-of-test functionality	
Technique:	Execute each use case, use-case flow, or function, using valid and invalid data, to verify the following: ● The expected results occur when valid data is used. ● The appropriate error or warning messages are displayed when invalid data is used. ● Each rule is properly applied.
Entry Criteria	● Application construction is completed. ● The test engineers are dedicated. ● Necessary devices, instruments, and other equipment are acquired. ● Test environment is prepared and the application is released to the test environment.
Completion Criteria:	● All the planned tests are performed. ● There are no show-stopping errors. ● All the errors of high priority and severity are fixed. ● The test results are evaluated, discussed and approved.
Special Considerations:	None.

4.2. Test Procedure

Test procedure assumes the next points:

- Reporting of found software bugs.

Various aspects of the tested software should be checked; this requires executing of different testing types.

The main testing types that would be executed:

- Functional Testing
- UI Testing
- Usability Testing
- Compatibility Testing (4 modern web browsers and devices)
- Regression testing
- Retesting (during the second round if needed)

It also will be checked how the software product is run on the browsers and devices that are supposed to support it, how it starts and stops, and how much time it needs to launch.

During this test round the next types of testing will NOT be applied:

- Security testing
- Load/Performance testing

4.3. Bug Reports

Bug reports are created in order to provide the development team and the project managers with exhaustive information about the discovered defects. They must be helpful in determining causes of the errors and correcting them.

Defect Severity can be classified into four categories:

- **Critical (blocker)** defects are the failure of the complete software system or of a critical subsystem, and no work or testing can be carried out after the occurrence of the defect. It also applies to data loss failures and with processes that leave inconsistent data stored in the database.
- **Major defects** (and crashes) are those which also cause failure of the entire or part of the system, but there are some processing alternatives which allow further operation of the system. It also applies to the system crashing, or aborting, during normal operation of a non-critical flow.
- **Minor defects** do not result in failure but cause the system to show incorrect, incomplete, or inconsistent results.
- **Trivial defects** are small errors that do not affect the functionality: typos, grammar mistakes, wrong terminology, etc.

The information that is indicated in each bug report:

- the software product name (SmartBreakReminder).
- version number of the software product;
- the browser on which the tests were performed.

Each report provides the next information about the defect:

- summary, which is short description of the problem;
- location of the defect in the software product;
- steps to reproduce the error;
- frequency of the defect occurrence;
- severity of the defect;
- additional information about the defect in the form of attached screenshots or video records.

Third party software will be used for reporting and maintaining discovered errors. The test team will log in that software all the defects that will be found during the testing process.

5. Resources

5.1. Tools

The following tools will be used for this project:

Name of process	Tool
Defect Tracking	Jira
Test Cases	TestRail
Screenshots / Video capture	Snagit
API Testing	Postman
Browser DevTools	Chrome DevTools

5.2. The list of the browsers

Name of the browser	Version
Chrome	Latest
FireFox	Latest
Edge	Latest
Safari	Latest

5.3. The list of the devices

Name of the device	OS
iPhone devices	All supported iOS versions
Android devices	All supported Android versions
Desktop (Windows)	Windows 10 / 11
Desktop (macOS)	macOS Ventura / Sonoma

6. The criteria of quality

The SmartBreakReminder product should operate in accordance with the requirements and the functional specification. The product should not contain critical and blocking defects in the final version of the project.

7. Testing Process Risks

The next issues may influence testing works:

- Changes and modifications of the software product that were not planned and discussed with the test team beforehand;
- Changes in the software requirements that were not discussed with the test team beforehand;
- Delays in correcting/fixing errors;
- Delays in delivering new builds to the test team;
- Limited availability of team members due to academic schedule constraints;
- Dependency on third-party libraries (MongoDB, Express, React, Node.js) which may have compatibility issues.

8. Test Team Expectations

The test team must be provided with valid, updated documents during the whole testing process.

All the required equipment, instruments, devices and software must be acquired and prepared before the beginning of the testing process.

All show-stopping errors must be corrected as soon as possible.

Release notes should be added to each software release to the test team. The note must explain which elements, functions and features were added to the program and how these additions affect the software.

The developers should correct all the errors in the software modules before releasing a new version.

9. Responsibilities of Test Team Members

Project Manager (V. Urbanskyi)

- Managing the whole testing process.
- Providing all the needed resources for the testing activities.
- Communication with stakeholders and the university supervisor.

QA Tech Lead / System Architect (D. Pavliuk)

- Managing the QA team from a technical perspective.
- Analyzing the tasks and distributing them between team members.
- Communicating with the client/supervisor team and discussing all issues, providing recommendations before an update or release.
- Experience in participation of different SDLC models including Agile and Scrum.
- Creating test documentation, including test cases, test plans, etc.
- Proposing best practices and tools for a project.
- Reviewing the system architecture for testability.

QA Engineer

- QA process / logging found errors into the approved bug tracking system (Jira).
- Writing and executing test cases in TestRail.
- Providing daily test execution reports.
- Retesting fixed defects.

10. Deliverables

- Test Plan (this document).
- Test Cases — functional test cases created in TestRail covering all scope areas.
- Bug Reports — defect reports logged in Jira with full details and attachments.
- Daily Status Reports — progress reports sent to the project manager each day.
- Final Test Report — summary of testing activities, defect statistics and recommendations.